

Reporte de Análisis Físico

Validación MC y Asimetría UMD/SD

Configuración de la Simulación

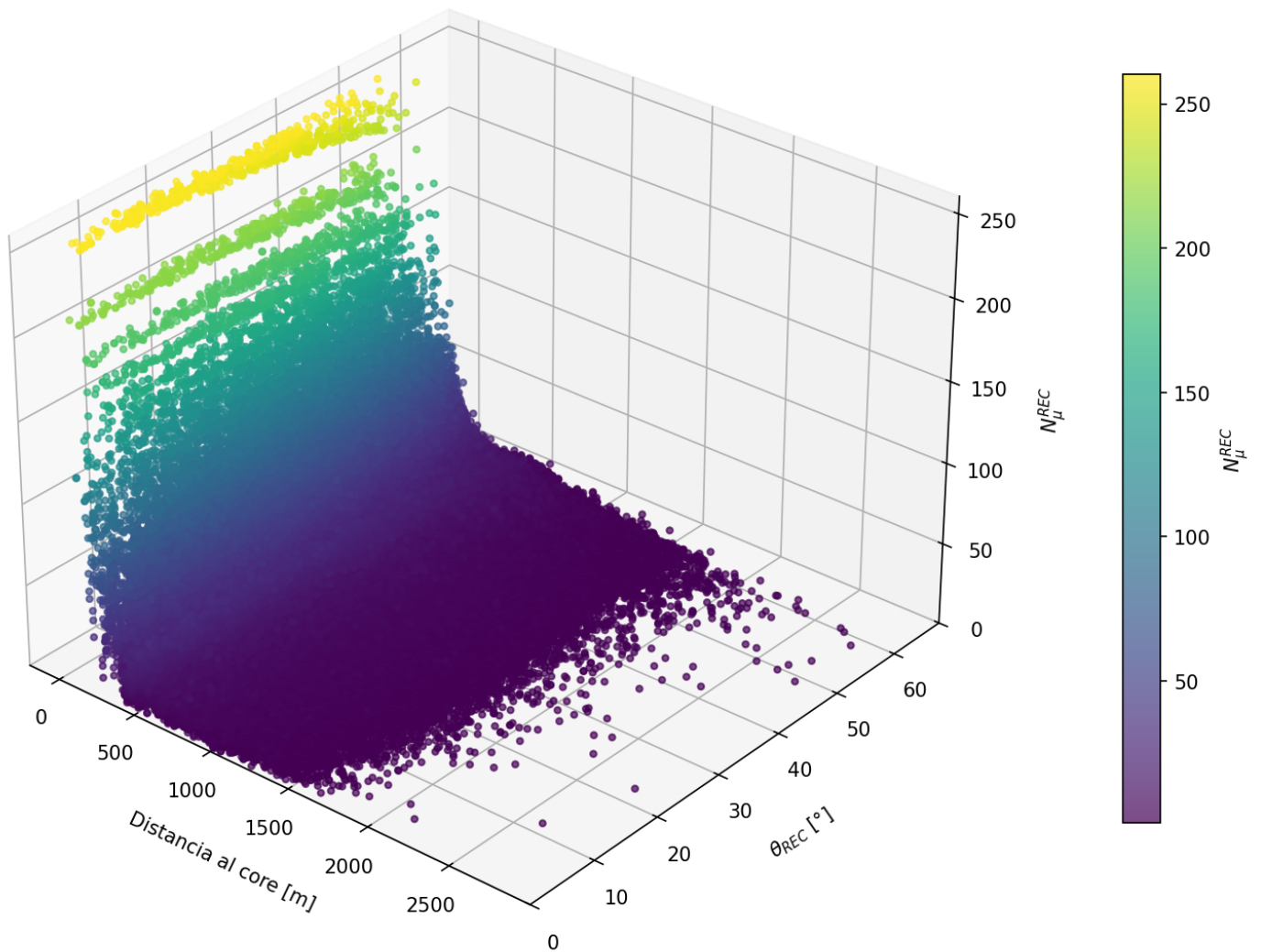
Parámetro	Valor
Modelo Hadrónico	SIBYLL 2.3d
Primario	Hierro
Rango Energía	17.5 - 18.0 log(eV)
Carpeta	
Eventos Totales	4,999
Módulos Totales	1,679,238

1. Limpieza de Datos

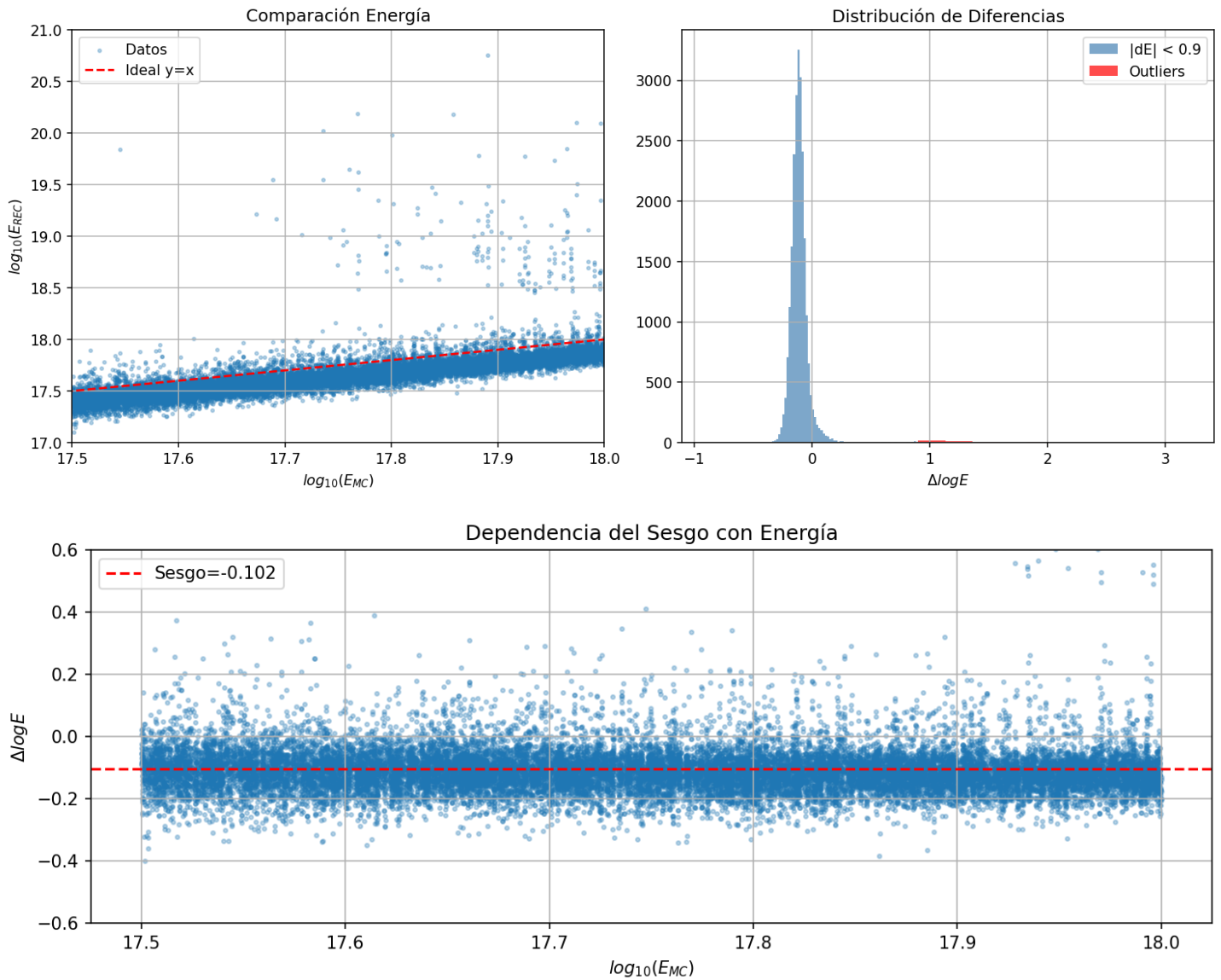
REPORTE DE LIMPIEZA

Métrica	Valor
Filas Iniciales	1679238
Filas Eliminadas	76509 (4.56%)
Filas Finales	1602729
Causa Principal	NaN en logE_REC (76509)

2. Visualización General

N_{μ}^{REC} vs distancia y θ 

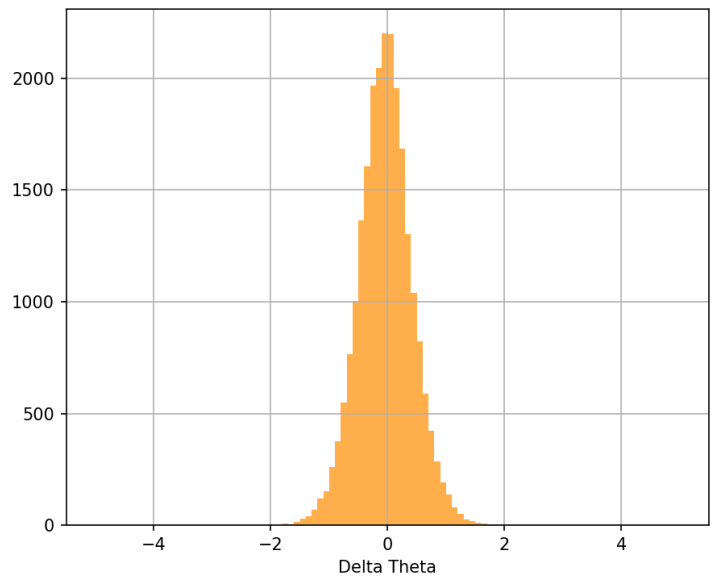
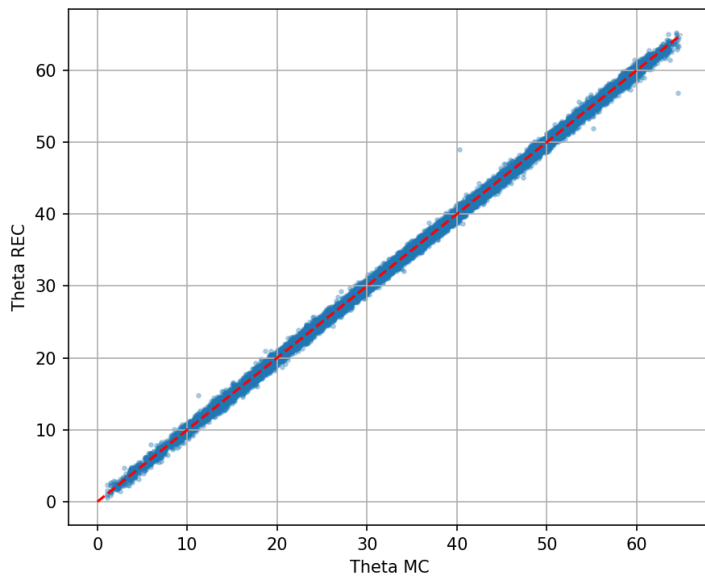
3. Resolución de Energía



Sesgo (Mean): -0.1025
 Resolución (Std): 0.1247
 Outliers (> 0.9): 88 (0.38%)

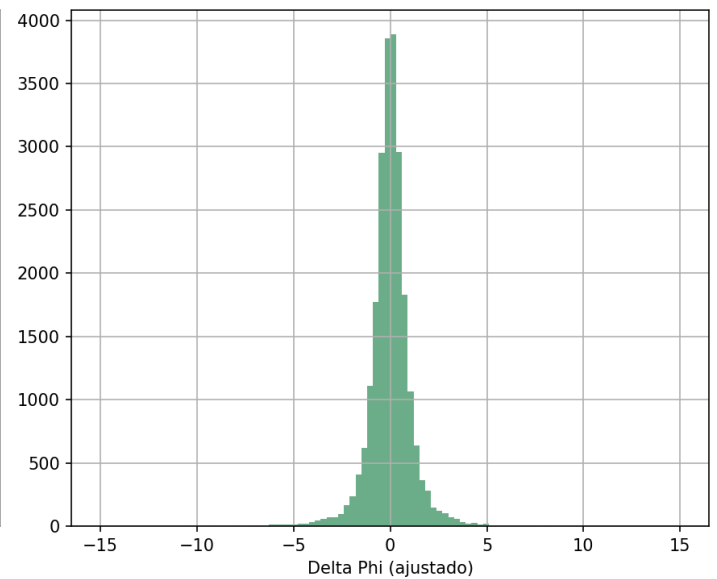
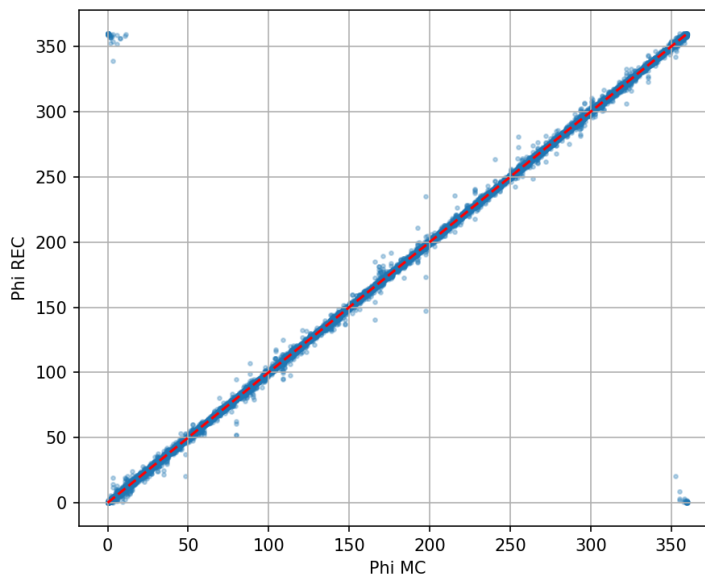
4. Resolución Angular

Theta (Cenital)



Sesgo: -0.0387 deg
Resolución: 0.4623 deg
Outliers ($> 10.0^\circ$): 0 (0.00%)

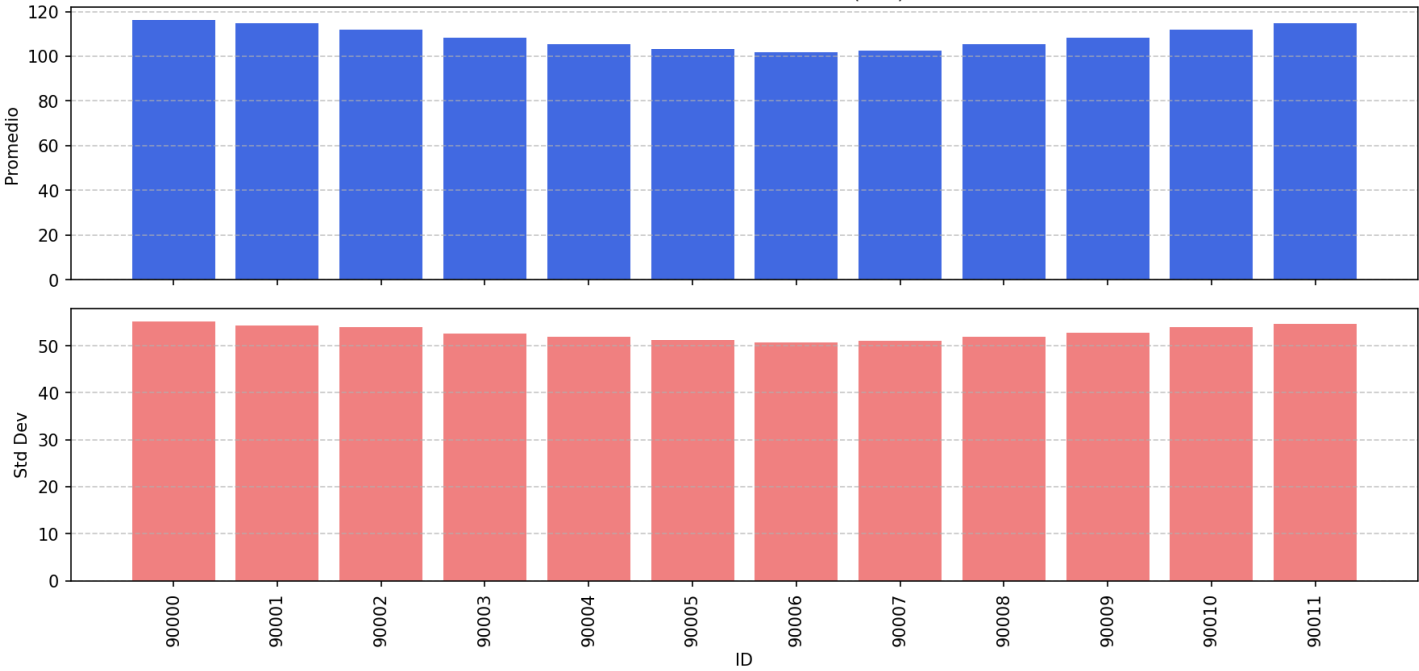
Phi (Azimutal)



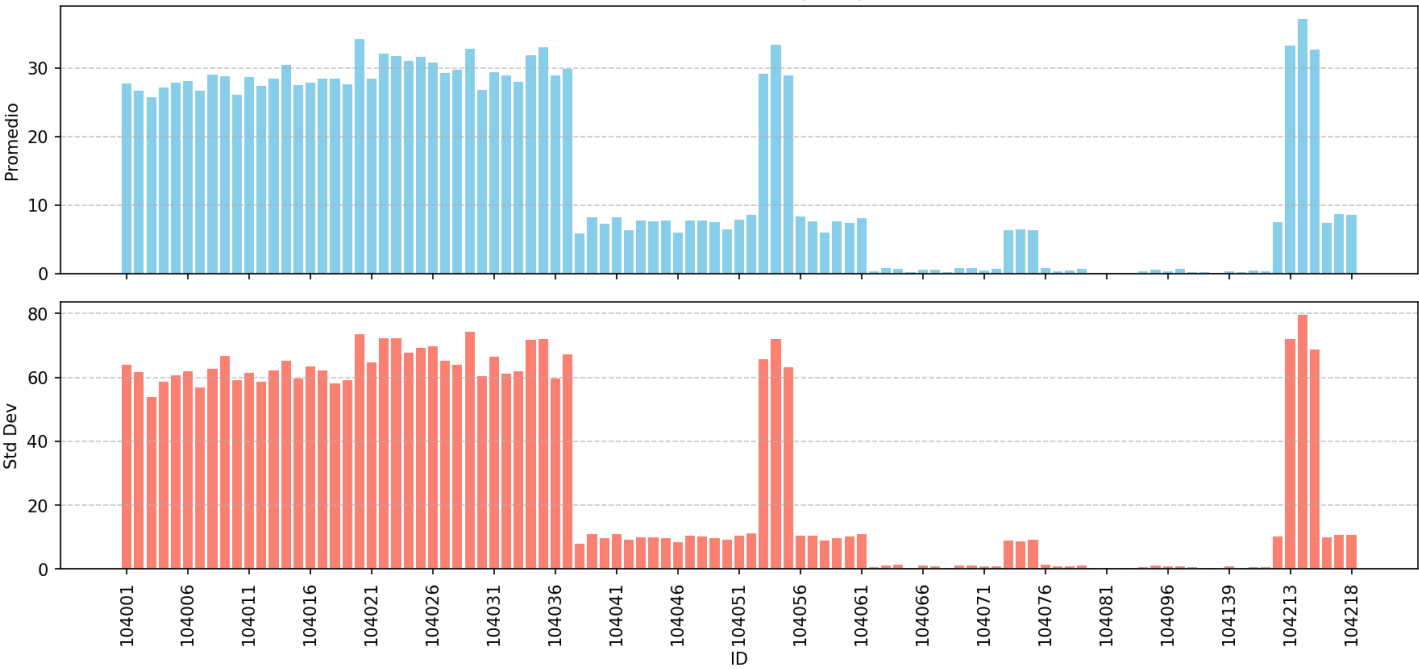
Sesgo (Median): 0.0030 deg
Resolución (MAD): 0.4831 deg
Outliers ($> 15.0^\circ$): 25 (0.11%)

5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

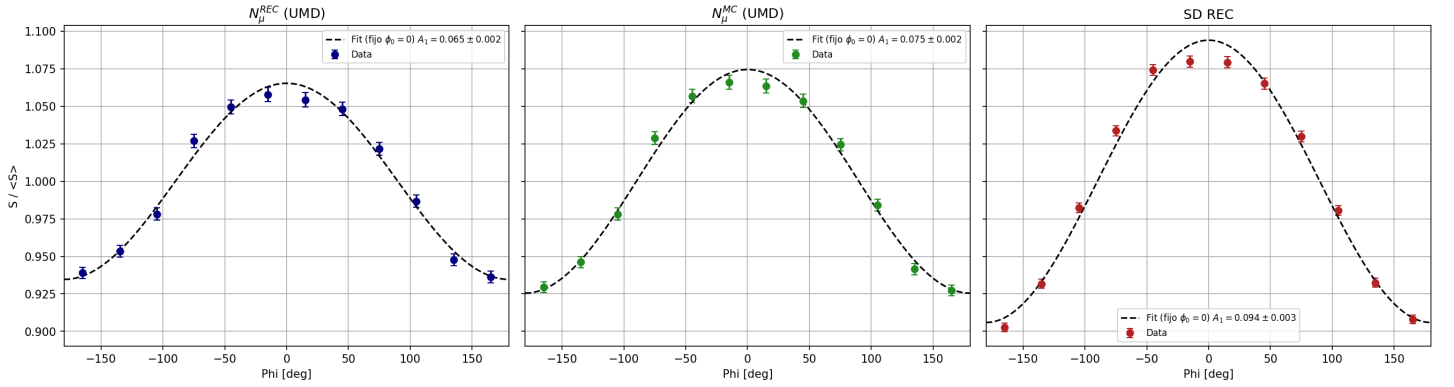
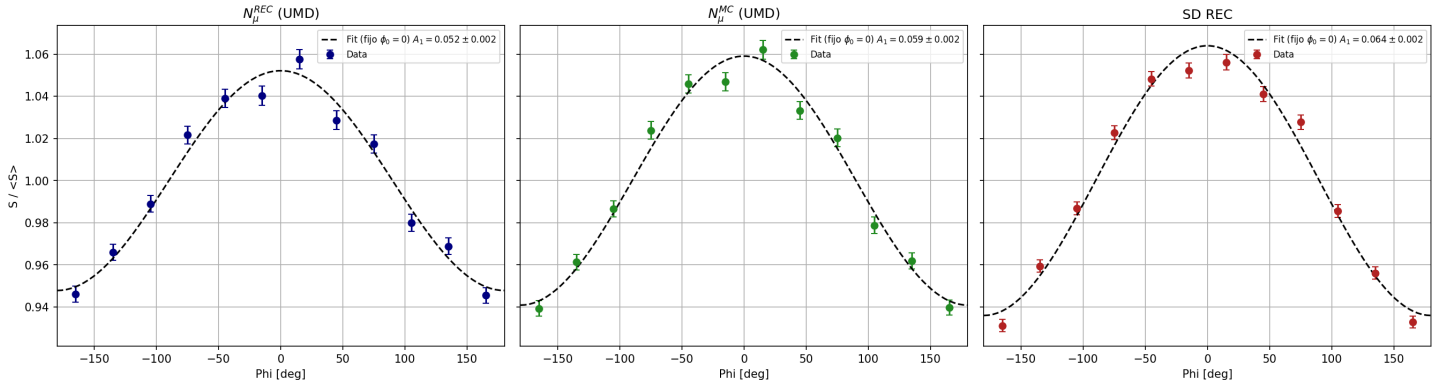
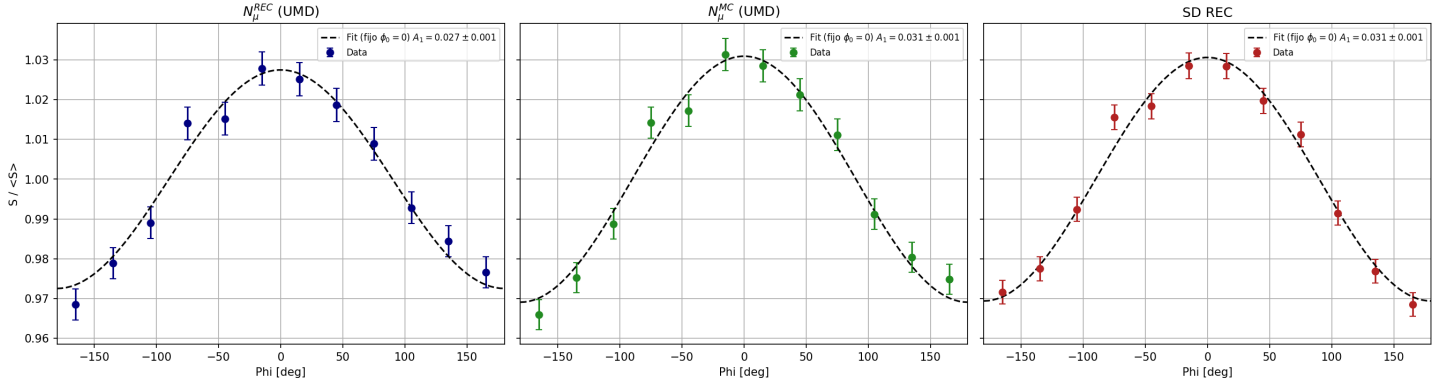


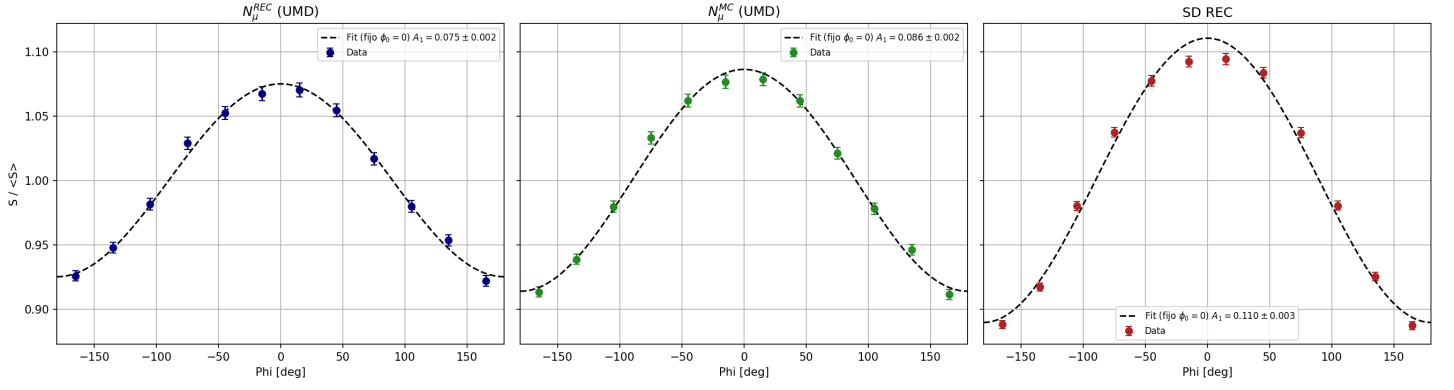
Uniformidad: Infill (104k)



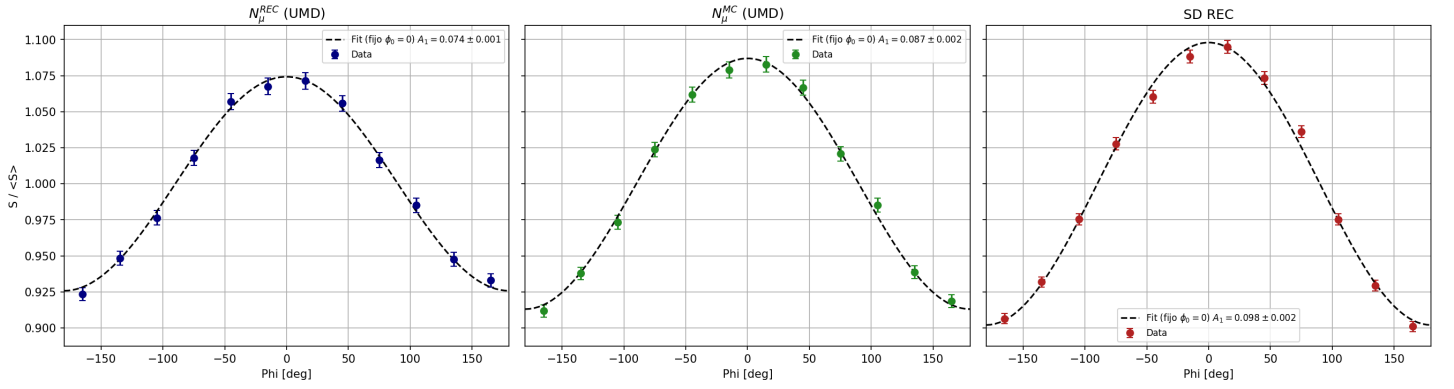
6. Asimetría Azimutal

Bin 0: Theta 0.5 - 21.7 deg

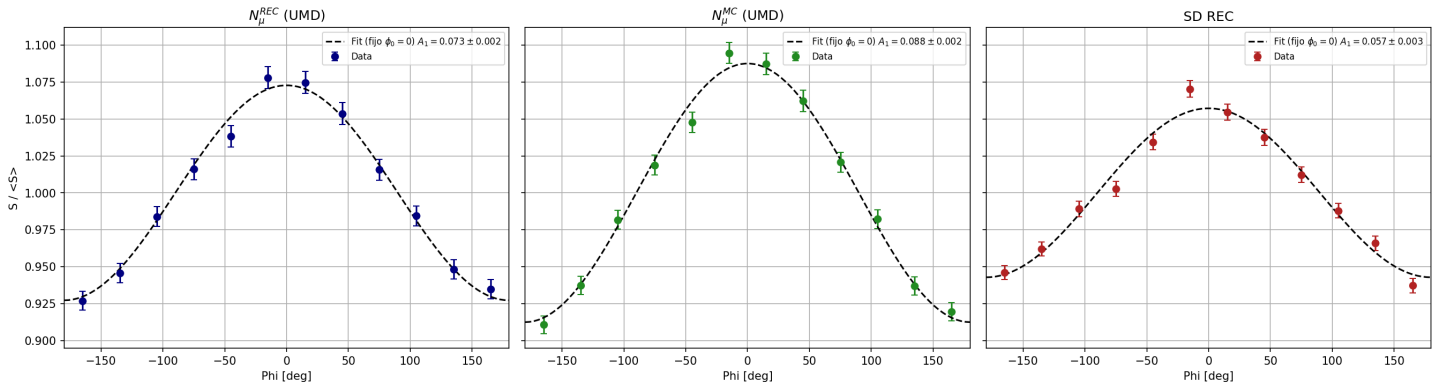




Bin 4: Theta 47.7 - 55.8 deg



Bin 5: Theta 55.8 - 65.0 deg



7. Resumen de Ajustes (Comparación)

Resultados Detallados: Amplitud A1 y Error

theta_range	A1_UMD_REC	Err_UMD_REC	A1_UMD_MC	Err_UMD_MC	A1_SD_REC	Err_SD_REC
0.5-21.7	0.0275	0.0015	0.0309	0.0015	0.0306	0.0012
21.7-31.6	0.0522	0.0024	0.0591	0.0024	0.0640	0.0025
31.6-39.9	0.0654	0.0021	0.0745	0.0020	0.0941	0.0027
39.9-47.7	0.0749	0.0018	0.0862	0.0020	0.1104	0.0032
47.7-55.8	0.0742	0.0015	0.0870	0.0015	0.0980	0.0019
55.8-65.0	0.0728	0.0022	0.0876	0.0024	0.0572	0.0028

